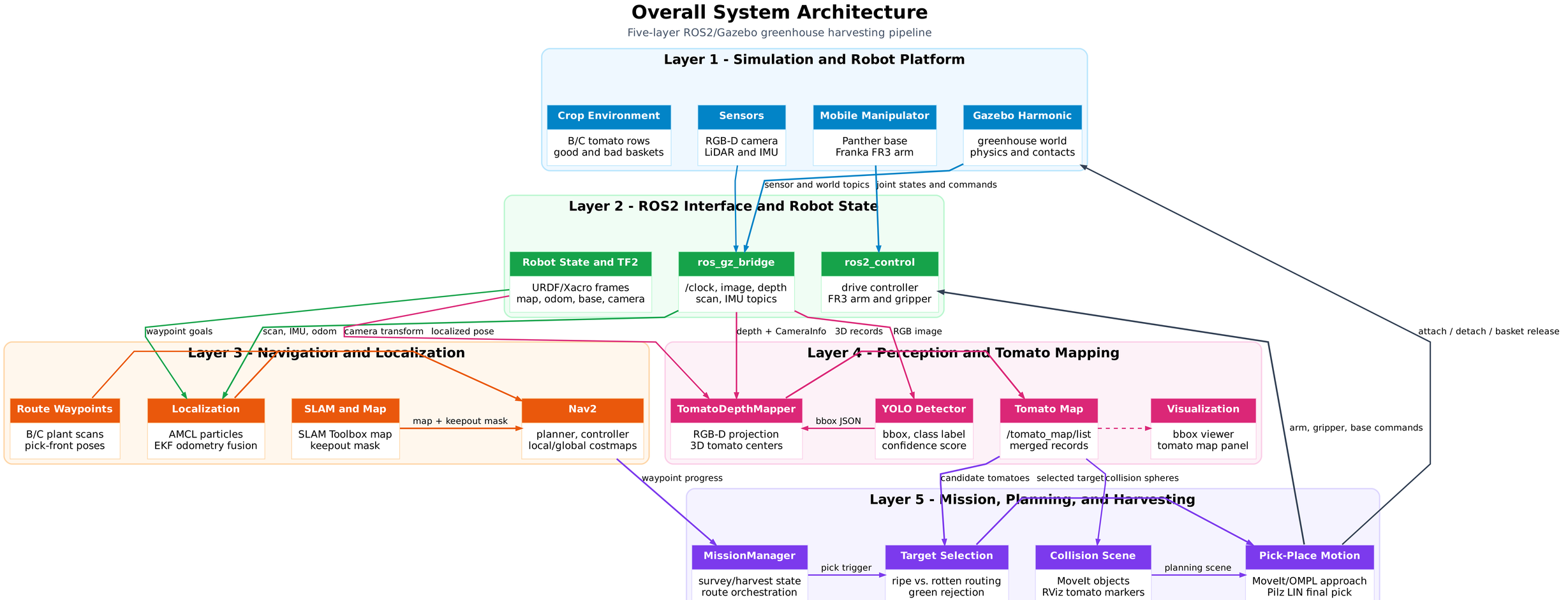
Overall System Architecture
Five-layer ROS2/Gazebo greenhouse harvesting pipeline
Layer 1 - Simulation and Robot Platform
Crop Environment
B/C tomato rows
good and bad baskets
Sensors
RGB-D camera
LiDAR and IMU
Mobile Manipulator
Panther base
Franka FR3 arm
Gazebo Harmonic
greenhouse world
physics and contacts
sensor and world topics
joint states and commands
Layer 2 - ROS2 Interface and Robot State
Robot State and TF2
URDF/Xacro frames
map, odom, base, camera
ros_gz_bridge
/clock, image, depth
scan, IMU topics
ros2_control
drive controller
FR3 arm and gripper
waypoint goals
scan, IMU, odom
camera transform
localized pose
depth + CameraInfo
3D records
RGB image
attach / detach / basket release
Layer 3 - Navigation and Localization
Route Waypoints
B/C plant scans
pick-front poses
Localization
AMCL particles
EKF odometry fusion
SLAM and Map
SLAM Toolbox map
keepout mask
map + keepout mask
Nav2
planner, controller
local/global costmaps
Layer 4 - Perception and Tomato Mapping
TomatoDepthMapper
RGB-D projection
3D tomato centers
bbox JSON
YOLO Detector
bbox, class label
confidence score
Tomato Map
/tomato_map/list
merged records
Visualization
bbox viewer
tomato map panel
arm, gripper, base commands
waypoint progress
candidate tomatoes
selected target
collision spheres
Layer 5 - Mission, Planning, and Harvesting
MissionManager
survey/harvest state
route orchestration
pick trigger
Target Selection
ripe vs. rotten routing
green rejection
Collision Scene
MoveIt objects
RViz tomato markers
planning scene
Pick-Place Motion
MoveIt/OMPL approach
Pilz LIN final pick